SAECOIN — MINING BENCHMARK PLAN

Document: Mining_Benchmark_Plan.md
Project: SAECOIN
Version: 1.0
Status: Draft — Implementation Pending
Consensus: Proof of Work (PoW)
Mining Targets: CPU and GPU
Primary Development Environment: Windows 11
Development PC RAM: 16 GB

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1. PURPOSE

This document defines the benchmarking strategy for SAECOIN mining software.

The objectives are to:

- Measure CPU mining performance.
- Measure GPU mining performance.
- Compare different hardware configurations.
- Measure electricity consumption.
- Monitor hardware temperatures.
- Evaluate mining stability.
- Measure mining efficiency.
- Identify hardware compatibility issues.
- Evaluate ASIC-resistance goals.

Benchmarking must begin only after the mining algorithm and consensus rules have been sufficiently specified.

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2. BENCHMARKING PRINCIPLES

All mining benchmarks must follow these principles:

1. Use the same mining algorithm when comparing hardware.
2. Record exact hardware and software configurations.
3. Record actual hashrate, not estimated marketing performance.
4. Measure electricity consumption separately from hashrate.
5. Monitor CPU and GPU temperatures.
6. Run tests long enough to detect performance instability.
7. Repeat tests to identify inconsistent results.
8. Never compare results from different algorithm configurations as if they were equivalent.

A higher hashrate does not automatically mean better mining efficiency.

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3. HARDWARE CATEGORIES

3.1 CPU Mining

Test different CPU categories:

- Entry-level desktop CPUs.
- Mid-range desktop CPUs.
- High-performance desktop CPUs.
- Laptop CPUs.
- Multi-core workstation CPUs.

Record:

- CPU model.
- Number of physical cores.
- Number of logical threads.
- Base frequency.
- Maximum observed frequency.
- CPU temperature.
- CPU utilization.
- RAM usage.
- Mining hashrate.
- Power consumption.

3.2 GPU Mining

Test:

- NVIDIA GPUs.
- AMD GPUs.
- Integrated GPUs, if supported.
- Different GPU memory capacities.
- Different GPU generations.

Record:

- GPU model.
- GPU architecture.
- VRAM capacity.
- Driver version.
- GPU utilization.
- GPU temperature.
- GPU power consumption.
- Mining hashrate.

- Memory usage.
- Stability.

GPU support must be implemented and tested independently for each supported hardware backend.

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4. BENCHMARK SOFTWARE

The benchmark environment should record:

Component| Information
Operating System| Windows 11 / Linux
CPU| Exact model
GPU| Exact model
RAM| Capacity and speed
Storage| SSD/HDD type
Mining Software| SAECOIN miner version
Node Software| SAECOIN node version
Mining Algorithm| Exact algorithm and version
Driver| GPU driver version
Compiler| Compiler and version
Build Type| Debug / Release

Use release builds for performance measurements.

Debug builds may be used for correctness testing but should not be treated as representative performance results.

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5. CPU BENCHMARK PROCEDURE

5.1 Preparation

1. Install the required mining software.
2. Start a local testnet node.
3. Confirm node synchronization.
4. Verify that mining work is valid.
5. Record the CPU configuration.
6. Close unnecessary background applications.

5.2 Test Configurations

Run separate tests using:

- One CPU thread.

- Two CPU threads.
- Half of available logical threads.
- All available logical threads.

Do not assume that using every thread produces the best efficiency.

5.3 Measurements

Record:

- Average hashrate.
- Peak hashrate.
- Minimum hashrate.
- CPU utilization.
- CPU temperature.
- System power consumption.
- RAM consumption.
- Mining stability.

5.4 Duration

Suggested test durations:

- Initial test: 5 minutes.
- Performance test: 30 minutes.
- Stability test: 2 hours.
- Extended test: 24 hours.

Long-duration testing should be performed only with suitable cooling and power monitoring.

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6. GPU BENCHMARK PROCEDURE

6.1 Preparation

1. Install compatible GPU drivers.
2. Confirm GPU detection.
3. Start a synchronized testnet node.
4. Verify mining algorithm compatibility.
5. Record GPU model and VRAM.
6. Check cooling and power conditions.

6.2 Measurements

Record:

- Average hashrate.

- Peak hashrate.
- GPU utilization.
- GPU temperature.
- GPU power consumption.
- VRAM usage.
- Driver stability.
- Invalid proof count.
- Mining interruptions.

6.3 GPU Stability Tests

Test:

- GPU startup.
- Mining for 30 minutes.
- Mining for 2 hours.
- Mining for 24 hours.
- Driver restart.
- GPU workload interruption.
- Recovery after node disconnection.

The miner must handle GPU failures without corrupting node or wallet data.

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7. CPU VS GPU COMPARISON

CPU and GPU results must be compared using the same algorithm and equivalent mining work.

Comparison metrics:

Metric| CPU| GPU
Hashrate| Measured| Measured
Power consumption| Measured| Measured
Temperature| Measured| Measured
Stability| Measured| Measured
Energy efficiency| Calculated| Calculated
Cost per hashrate| Calculated| Calculated

The comparison must consider total system electricity consumption as well as the incremental power used by the mining hardware.

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8. ELECTRICITY CONSUMPTION

Mining electricity consumption must be measured using a reliable power meter where practical.

Record:

- Idle system power.
- Mining system power.
- Mining duration.
- Average power consumption.
- Total energy consumed.

Calculation:

Energy (kWh) = Average Power (W) × Hours ÷ 1000

Example:

A mining system consuming 200 watts for 10 hours:

200 × 10 ÷ 1000 = 2 kWh

Electricity cost:

Electricity Cost = Energy Consumed × Electricity Rate

Use the actual electricity tariff applicable to the test location.

Do not estimate profitability using hashrate alone.

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9. MINING EFFICIENCY

Calculate:

Hashrate per Watt = Hashrate ÷ Power Consumption

Example:

- Hashrate: 1,000 H/s
- Power consumption: 100 W

Efficiency:

10 H/s per watt.

For different hardware, report the units clearly.

A benchmark report must identify whether power consumption represents:

- CPU or GPU only.
- Entire computer.
- Entire mining system including cooling.

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10. TEMPERATURE AND THERMAL STABILITY

Monitor:

- CPU temperature.
- GPU temperature.
- GPU hotspot temperature where available.
- Fan speed.
- Thermal throttling.
- Ambient temperature.

Test objectives:

- No unsafe temperature behavior.
- No repeated thermal throttling.
- No unexpected shutdowns.
- Stable hashrate over time.

Use manufacturer-recommended temperature limits for the specific hardware.

Do not disable thermal protection mechanisms to increase mining performance.

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11. ASIC-RESISTANCE EVALUATION

ASIC resistance is a design objective, not a guarantee.

The benchmark process should investigate:

- CPU performance.
- GPU performance.
- Memory requirements.
- Parallel processing behavior.
- Hardware specialization advantages.
- Potential ASIC implementation feasibility.

Compare the relative performance of CPUs, GPUs, and specialized hardware if such hardware becomes available.

Do not claim SAECOIN is ASIC-resistant solely because CPU and GPU mining work.

The algorithm requires independent technical analysis.

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12. MULTI-MINER TESTING

Test multiple miners operating simultaneously.

Scenarios:

1. Two CPU miners.
2. Two GPU miners.
3. CPU and GPU miners together.
4. Multiple independent mining computers.
5. Miners with different performance levels.

Measure:

- Combined hashrate.
- Block production.
- Stale work.
- Invalid proofs.
- Network propagation.
- Miner stability.

All miners must follow the same consensus rules.

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13. NETWORK-LEVEL MINING BENCHMARKS

Mining performance must also be evaluated at the network level.

Record:

- Total estimated network hashrate.
- Average block interval.
- Difficulty changes.
- Block propagation time.
- Orphan/stale block rate.
- Miner distribution.
- Hashrate concentration.

Test sudden changes in mining participation.

Examples:

- 50% of miners disconnect.
- A powerful miner joins.
- A powerful miner leaves.
- Network latency increases.
- Network connectivity becomes unstable.

The purpose is to evaluate whether the network remains predictable and secure.

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14. ASIC AND MINING CENTRALIZATION RISKS

Monitor whether a particular hardware category gains a disproportionate advantage.

Evaluate:

- CPU-to-GPU performance ratio.
- GPU-to-specialized-hardware performance ratio.
- Cost of entry.
- Memory requirements.
- Power efficiency.
- Mining concentration.

A practical mining ecosystem should avoid unnecessary barriers to participation, but no algorithm can guarantee equal rewards for all hardware.

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15. BENCHMARK RESULT FORMAT

Every benchmark report should contain:

Hardware Information

- CPU model:
- GPU model:
- RAM:
- Storage:
- Operating system:
- Driver version:

Software Information

- SAECOIN node version:
- Miner version:
- Algorithm version:
- Build type:

- Network:

Performance Results

- Average hashrate:
- Peak hashrate:
- Minimum hashrate:
- CPU utilization:
- GPU utilization:
- Average temperature:
- Maximum temperature:
- Average power:
- Energy consumed:
- Invalid proofs:
- Stale work:

Stability Results

- Test duration:
- Crashes:
- Driver failures:
- Mining interruptions:
- Thermal throttling:
- Final result:

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16. BENCHMARK ACCEPTANCE CRITERIA

A mining implementation should pass the following checks:

1. Produces valid proofs.
2. Produces consistent results.
3. Does not submit invalid blocks.
4. Reports hashrate accurately.
5. Handles node disconnections.
6. Handles hardware failures safely.
7. Maintains stable operation.
8. Does not bypass thermal protections.
9. Uses resources within documented limits.
10. Produces reproducible benchmark results.

Performance targets must be established only after the final algorithm is selected and initial hardware measurements are available.

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17. DEVELOPMENT PC BASELINE

The primary development computer has:

- Operating System: Windows 11.
- RAM: 16 GB.

This machine can be used for:

- CPU mining tests.
- Basic GPU compatibility testing, if supported hardware exists.
- Local testnet experiments.
- Benchmark software development.
- Short-duration performance tests.

It should not automatically be assumed suitable for large-scale mining or long-duration stress testing.

Actual CPU and GPU models must be recorded before making performance predictions.

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18. FINAL PRINCIPLE

SAECOIN mining must be evaluated using measured performance, energy consumption, stability, and network behavior.

A miner that reports a high hashrate but produces invalid work is not successful.

A miner that performs well for five minutes but crashes after several hours is not production-ready.

Mining performance must never take priority over consensus correctness and network security.

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